

Design of a Bipedal Walker Controlled by a CPG Network in Animatlab^{*}

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Abstract. This work details a neuromechanical model of a bipedal robot based off simplified anatomy of the human lower body. The controller consists of a neural network that replicates central pattern generators (CPGs) that take the form of rhythm generators (RGs), pattern formation (PF) networks, and sensorimotor networks. The current model is unable to walk on it's own, however this work demonstrates that biologically similar gait patterns can be generated through the use of CPGs.

Keywords: CPG · Biped · Animatlab

1 Introduction

Biomimetic robots attempt to replicate human biology to further research in both robotics and anatomy. By replicating both the biomechanical properties and neural systems of a human, researchers can formulate hypotheses and test them in a controlled environment.

These robotics allow for continued research into CPGs and how humans use them to generate smooth and continuous gait while encountering perturbations. Literature on CPGs is numerous [2] and their implementation on robotics is an active area of research [4]. However, most robot models that use CPGs use simplified models of the neural networks to drive actuation.

This work takes that research and creates a model of a bipedal walker in the neuromechanical modeling software Animatlab [1]. The Animatlab environment allows researchers to construct physical models that are controlled by a synthetic neural network. The simulation detailed in this work looks to creating a model to be used on biomimetic bipedal robots.

2 Methods

The neuromechanical model was developed in Animatlab. Animatlab is a 3D graphical environment that allows user to construct biomechanical systems that are actuated by synthetic neural systems (SNS). A bipedal humanoid biomechanical model was developed in Animatlab, whose physical structure is based

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off of the work by Li et al. [6]. In their work, Li creates a SNS with spiking neurons that form both a rhythm generator (RG) network and pattern formation (PF) network that control the muscles, but lacks afferent feedback from the muscles. The SNS developed here is based off of the SNS created by Deng et al. [3]. Their network has a RG network, PF network, sensorimotor network, and afferent feedback to modulate these networks.

2.1 Biomechanical Model

The biomechanical model created simulates the simplified anatomy of the human lower body. This model consists of nine bodies: pelvis, thighs, shanks, feet, and toes (Figure 1a). Hinge joints between segments allow for rotation with capabilities similar to human anatomy. The model is designed to only perform gait in the sagittal plane and therefore does not require joints with more degrees of freedom.

Attached to the body of the model are Hill muscle model actuators. Antagonistic muscles about each joint create rotation. The only joint not to contain a pair of antagonistic muscles is the toe joint, which contains a spring instead to increase compliance of the foot and to conserve energy when walking (Figure 1b).

2.2 Synthetic Neural Network

The synthetic neural network is comprised of nonspiking neurons. The neuron dynamics are derived from a leaky-integrator neuron model, which can be described by the differential equations:

$$C_m \frac{dV}{dt} = I_{leak} + I_{syn} + I_{Na} + I_{app} \quad (1)$$

where C_m is the membrane conductance, V is the membrane voltage, I_{leak} is the leak current, I_{syn} is the synaptic current, and I_{app} is an applied current. Each current is governed by it's own equation:

$$I_{leak} = G_m(E_r - V) \quad (2)$$

$$I_{syn} = \sum_{i=1}^n G_{syn,i}(E_{syn,i} - V) \quad (3)$$

$$I_{Na} = G_{Na} \cdot m \cdot h(E_{Na} - V) \quad (4)$$

where E_r is the reversal potential, $G_{syn,i}$ is the incoming synaptic potential from the i th synapse, and E_{Na} is the sodium channel potential. The m and h parameters are the activation parameter and deactivation parameter of the sodium

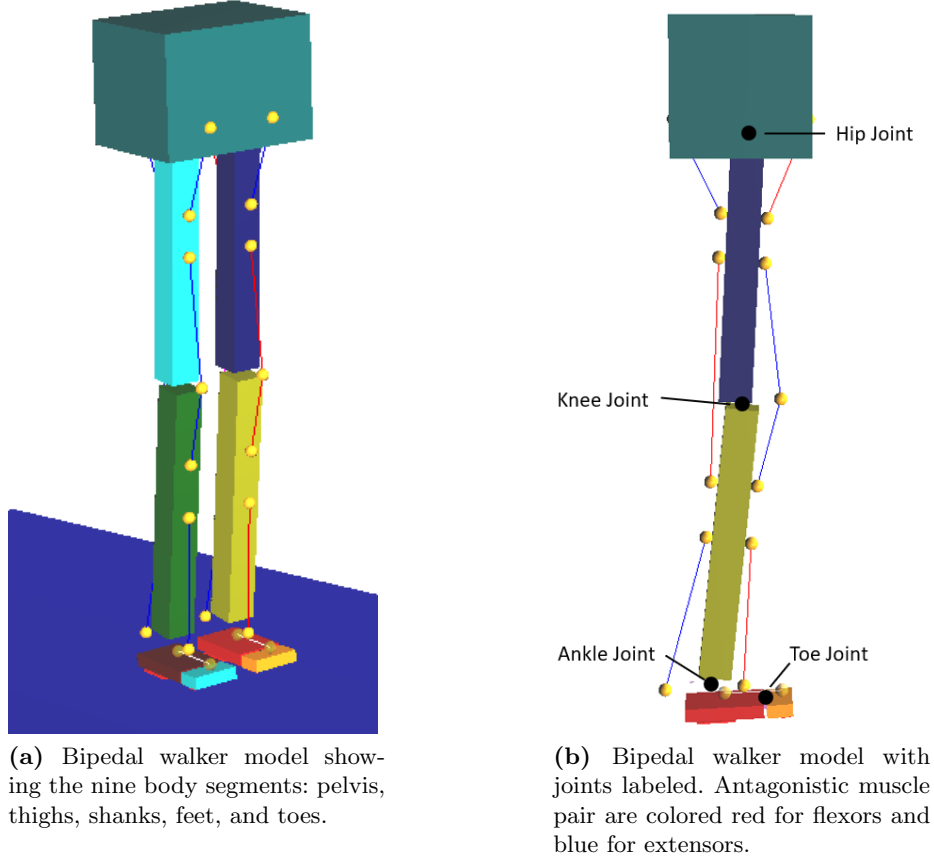


Fig. 1: Images of the bipedal walker model.

channel, respectively, and can be calculated by knowing their time constants, τ_m and τ_h :

$$\frac{dm}{dt} = \frac{m_\infty - m}{\tau_m} \quad (5)$$

$$\frac{dh}{dt} = \frac{h_\infty - h}{\tau_h} \quad (6)$$

Both m_∞ and h_∞ can be calculated from the same equation, substituting in the appropriate parameters into the p variables:

$$p_\infty = \frac{1}{1 + A_p \cdot e^{S_p(V-E_p)}} \quad (7)$$

where A_p is the factor, S_p is the slope, and E_p is the reversal potential. The synaptic conductance for a specific synapse, $G_{syn,i}$ is treated as a linear piece wise function governed by:

$$G_{syn,i} = \begin{cases} 0 & V_{pre} < E_{lo} \\ g_{syn} \frac{V_{pre} - E_{lo}}{E_{hi} - E_{lo}} & E_{lo} \leq V_{pre} < E_{hi} \\ g_{syn} & E_{hi} \leq V_{pre} \end{cases} \quad (8)$$

where V_{pre} is the presynaptic neuron voltage and g_{syn} , E_{lo} , and E_{hi} represent the slope, lower bound, and upper bound of the piece wise function. Values for all of these parameters used in the model can be found in Table 1 and 2.

Table 1: Neural Parameters

Neuron	C_m (nF)	G_m (μS)	E_r (mV)	G_{Na} (μS)	A_h	S_h	E_h (mV)	τ_h (ms)	A_m	S_m	E_m (mV)	τ_m (ms)
RG	5	1	-60	1.5	0.5	-0.6	-60	350	1	0.2	-40	2
PF	5	1	-60	1.5	0.5	-0.6	-60	350	1	0.2	-40	2
MN	5	1	-100	0	-	-	-	-	-	-	-	-
IN	5	1	-60	0	-	-	-	-	-	-	-	-

Table 2: Synapse Parameters

Synapse	g_{syn} (μS)	E_s (mV)	E_{lo} (mV)	E_{hi} mV
RG to IN	2.749	-40	-60	-25
IN to RG	2.749	-70	-60	-25
RG to RG	0.1	-40	-65	-40
PF to IN	2.749	-40	-60	-25
IN to PF	2.749	-70	-60	-25
RG to PF	0.1	-40	-60	-40
PF to MN	-	-10	-60	-50
PF to Ia	0.5	-40	-60	-40
Ia to Ia	0.5	-70	-60	-40
Ia to MN	2	-100	-60	-40
MN to RE	0.5	-40	-100	-10
RE to RE	0.5	-70	-60	-40
RE to MN	0.5	-100	-60	-40
RE to Ia	0.5	-70	-60	-40

CPGs are created from four coupled nonspiking neurons that oscillate periods of activity. Two neurons serve as outputs to different networks. When one output neuron is driven high the other is driven low. Two integrator neurons are included

in the network to drive this oscillation of activation and inhibition. These CPG networks take the form of a rhythm generator network and a pattern formation network [7].

RG networks maintain the timing of the system. The RG network is the highest network in the hierarchical structure and is based off of the findings of Zhang et al. [8]. Two output neurons are connected through mutual excitation and are attached via inhibitory synapses to interneurons. The structure of the used RG network can be seen in Figure 2a.

PF networks help reshape the timing signals coming from the RG layer and dictate the activity of the sensorimotor layer. The PF layer also controls muscle synergies between the knee and ankle muscles, activating them with similar timing.

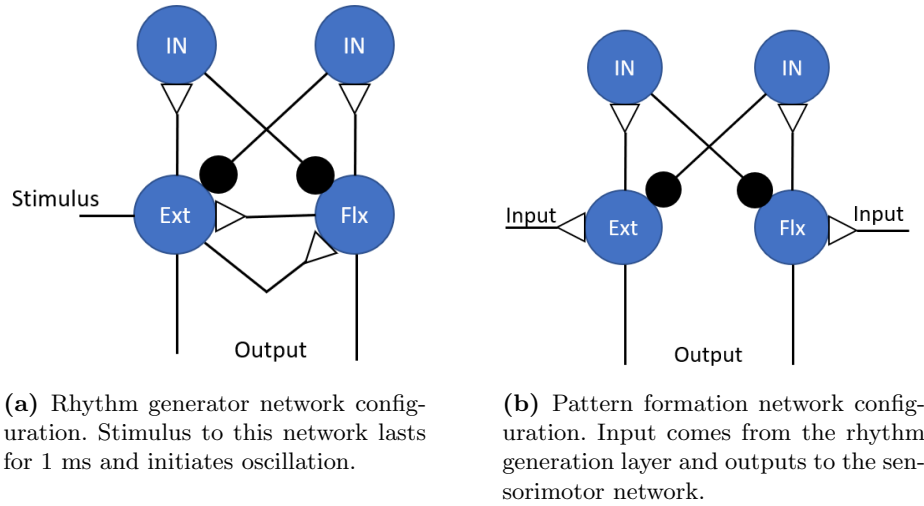


Fig. 2: Example configuration of the rhythm generation and pattern formation networks.

A third simplified CPG network is included in the sensorimotor network and oscillates activity of motor neurons that drive the muscles. This network is made of two Renshall cells coupled by inhibitory synapses. The Renshall cells provide feedback, along with Ia signals, Ib signals, and PF network output, to the motor neurons. These motor neurons output their activity to muscles to drive contraction and relaxation. Figure 3 details the connections of the used sensorimotor network.

Muscles provide feedback through afferent receptors. Both Ia and Ib signals are used in the feedback for the sensorimotor network. The Ia signals cause an interneuron to discharge at a rate that is proportional to the velocity of stretch in the muscle. The Ib signals cause an interneuron to discharge at a rate

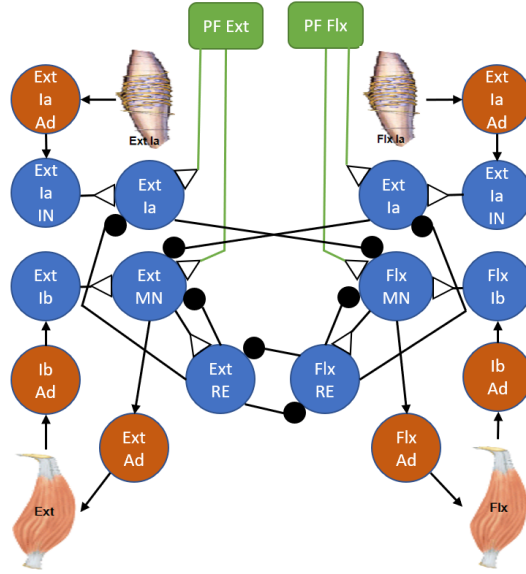


Fig. 3: Example configuration of a sensorimotor network, who receives input from the pattern formation layer.

proportional to the tension in the muscle. These feedback signals help reinforce the step and swing in the model as well as signaling when the leg should begin switching between the two modes.

3 Results

The oscillating signals from the different neural networks create a stepping pattern in the bipedal biomechanical model. The RG, PF, and motor neurons oscillate their levels of activation once the simulation is initiated. Figure 4 shows that each couple neuron pair switches its voltage with a frequency of 4 Hz. The output from these neurons drive the activation of the muscles on the model. As one muscle in an antagonistic muscle pair begins to contract, the other extends. This creates rotation about the joint (Figure 5). These figures show the results from the left sided limbs, which are mirrored for the right side limbs. The model is also suspended in the air. When the model is lowered to a platform below it, it is unable to use the gait patterns to generate forward motion.

4 Discussion

The results show that a neural network comprised of a rhythm generator layer, pattern formation layer, and sensorimotor layer can be used to actuate and drive a bipedal model in a manner similar to gait. All layers of the network activate

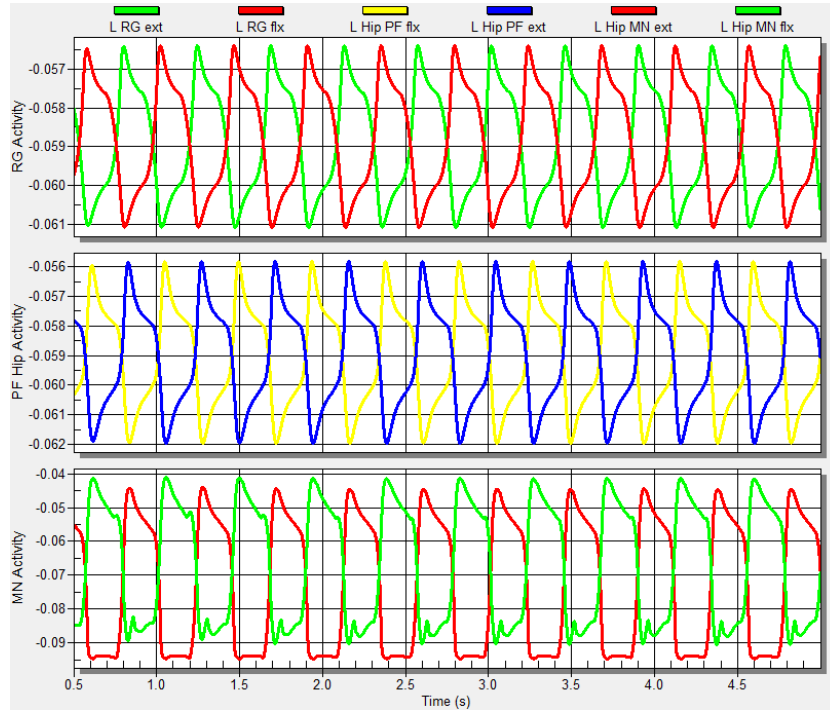


Fig. 4: Output from the rhythm generation layer (top), pattern formation layer (middle), and motor neurons in the sensorimotor layer (bottom).

at frequencies that are in phase with one another and the motor neuron outputs properly actuate the muscles that drive the physical model.

The timing of the oscillation is faster than typical human gait, which is approximately 1 Hz [5]. The current model has a gait frequency of 4 Hz. This quick timing frequency can be adjusted by altering the activation and deactivation values for the sodium channels.

As the model moves, all joints move to their maximum and minimum positions. This abrupt collision in the model can be corrected by adjusting the stimulus tension values for the individual muscles. Muscles are stimulated by the motor neurons and output a tension that is dictated by a polynomial curve in Animatlab. By adjusting the values of this polynomial curve, the high tension values of the muscles can be lowered so that the joints don't reach their maximum and minimum values as quickly.

5 Conclusion

This work demonstrates the construction of a bipedal walker that is controlled with a 2 layer CPG hierarchy. The rhythm generator CPG network, located at

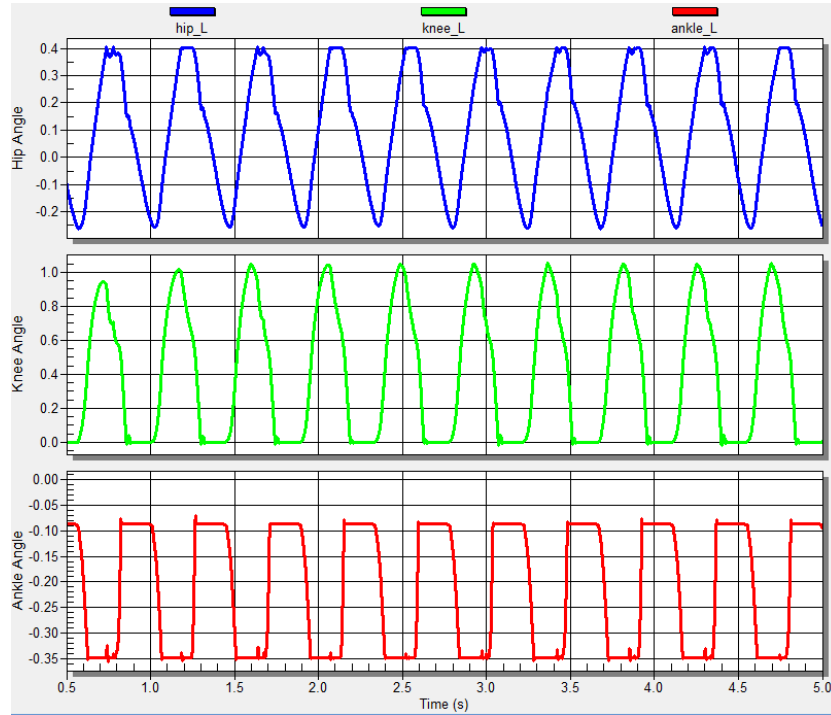


Fig. 5: Angle of the body joints as they are stimulated: hip (top), knee (middle), and ankle (bottom).

the top of the network hierarchy, is initiated by a stimulus of 10 nA for a duration of 1 ms, and creates a timing system for the rest of the networks. The pattern formation networks take in the output from the rhythm generator to create step timing frequency for the sensory motor networks. The sensory motor networks take in the pattern formation output as well as afferent signals from the muscles to modulate step timing. The signals from the sensory motor network drive pair of antagonistic muscles about a hip, knee, and ankle joint. Muscle tension and velocity feedback into the sensory motor network.

The model is able to generate gait like stepping motions, however when lowered to the ground, it is unable to walk. Walking can be generated with further tuning of neural and muscle parameters, as well as implementation of more feedback systems. By adjusting the activation and deactivation of the sodium channel parameters, the model will more closely resemble the kinematics of human gait, which will aid it while walking. By implementing more feedback systems, such as heel and toe contact, the model will be able to modulate its stepping motion. Contact sensors will help the neural network know when swing and stance are necessary, by reinforcing stance when the model is supporting all of its weight on one leg, while then reinforcing swing motions as the force on the contact sensor begins to decrease.

The model is a considerable step towards building biomimetic humanoid robots. By studying and analysing the motion that the model is capable of in simulation, similar systems can be physically built.

References

1. Cofer, D., Cymbalyuk, G., Reid, J., Zhu, Y., Heitler, W.J., Edwards, D.H.: AnimatLab: A 3D graphics environment for neuromechanical simulations. *Journal of Neuroscience Methods* 187(2), 280–288 (Mar 2010), <https://linkinghub.elsevier.com/retrieve/pii/S0165027010000087>
2. Danner, S.M., Shevtsova, N.A., Frigon, A., Rybak, I.A.: Computational modeling of spinal circuits controlling limb coordination and gaits in quadrupeds. *eLife* 6, e31050 (Nov 2017), <https://elifesciences.org/articles/31050>
3. Deng, K., Szczecinski, N.S., Arnold, D., Andrada, E., Fischer, M.S., Quinn, R.D., Hunt, A.J.: Neuromechanical Model of Rat Hindlimb Walking with Two-Layer CPGs. *Biomimetics* 4(1), 21 (Mar 2019), <https://www.mdpi.com/2313-7673/4/1/21>
4. Ijspeert, A.J.: Central pattern generators for locomotion control in animals and robots: a review. *Neural networks : the official journal of the International Neural Network Society* 21(4), 642–653 (May 2008), <http://www.ncbi.nlm.nih.gov/pubmed/18555958>
5. Kadaba, M.P., Ramakrishnan, H.K., Wootten, M.E.: Measurement of lower extremity kinematics during level walking. *Journal of Orthopaedic Research* 8(3), 383–392 (May 1990), <http://doi.wiley.com/10.1002/jor.1100080310>
6. Li, W., Szczecinski, N.S., Quinn, R.D.: A neural network with central pattern generators entrained by sensory feedback controls walking of a bipedal model. *Bioinspiration & Biomimetics* 12(6), 065002 (Oct 2017), <https://doi.org/10.1088%2F1748-3190%2Faa8290>
7. McCrea, D.A., Rybak, I.A.: Organization of mammalian locomotor rhythm and pattern generation. *Brain Research Reviews* 57(1), 134–146 (Jan 2008), <https://linkinghub.elsevier.com/retrieve/pii/S0165017307001798>
8. Zhang, J., Lanuza, G., Britz, O., Wang, Z., Siembab, V., Zhang, Y., Velasquez, T., Alvarez, F., Frank, E., Goulding, M.: V1 and V2b Interneurons Secure the Alternating Flexor-Extensor Motor Activity Mice Require for Limbed Locomotion. *Neuron* 82(1), 138–150 (Apr 2014), <https://linkinghub.elsevier.com/retrieve/pii/S0896627314001111>